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STATUS REPORT

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AUTH: ROBERT J. FRIEND, MAJOR USAF

By Robert J. Friend, Major
Signature and Grade

Date 9 SEPT 1960

PROJECT GRUDGE - REPORT NO.4

PROJECT NO. 10073

29 FEBRUARY 1952

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AUTH: ROBERT J. FRIEND, MAJOR, USAF

By Robert J. Friend, Major
Signature and Grade

Date 9 SEPT 1960

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DATE 13 Mar 52

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AUTH: ROBERT J. FRIEND, MAJOR USAF
By Robert J. Friend, Major
Signature and Grade

Date 7 SEPT 1960

This report is the fourth of a series of monthly status reports of Project Grudge. Each report will be written on or near the last day of the month and will contain a list of all incidents reported during the month covered by the report. The reports that are considered to be outstanding will be summarized in the appendices of the report so that more details can be presented. The overall status of the project will also be presented.

Additional information may be obtained on any incident by directing requests to Chief, Air Technical Intelligence Center, Attention: ATIAA-2c, Wright-Patterson Air Force Base, Dayton, Ohio.

The security classification of this report has been raised from Confidential to Secret due to the inclosure of reports of unidentified aerial objects which were classified Secret by the originating agency.

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STATUS OF PROJECT GRUDGE

I. OVERALL STATUS

A. Files

All of the material in ATIC that relates to sightings of unidentified aerial objects has now been filed and cross-indexed. Over 600 reports have been cross-indexed under the main divisions of:

1. Time of Sighting
2. Shape
3. Size
4. Course
5. Number of Objects Seen
6. Sounds
7. Date
8. Location
9. Occupation of Source
10. Color
11. Apparent Speed
12. Apparent Altitude
13. Length of Time Observed
14. Maneuvers
15. Conclusions

In many instances it has been difficult to establish sub-divisions due to the great variety of descriptions. In these instances, certain broad categories were established.

B. Location of Additional Files

The D/I Library files were searched during the past month and approximately 50 additional incidents were located. Copies of these have been requested. It is believed that the ATIC file on unidentified aerial objects now contains a large majority of all incidents reported to the Air Force since 1947.

C. Directives for Reporting Incidents

A new proposed directive for reporting sightings of unidentified aerial objects has been sent to the D/I for approval and distribution. This directive will replace existing directives and provide more expeditious channels for reporting sightings.

D. Project Twinkle

This Center has been receiving a number of reports on the phenomena which has been termed "green fireballs". This specific phenomena has been

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investigated by the AF Cambridge Research Laboratories under the title of "Project Twinkle". The Cambridge Laboratory has discontinued the project and the conclusions were indefinite.

This phenomena which has received some publicity in the past four months is reported to be similar to a large meteor in some respects. They are green in color, have a flat trajectory, appear to be much lower than an average meteor, and are silent.

II. REPORTS OF SPECIFIC INCIDENTS

A. Inclosed Summary List of Incidents

The inclosed list of incidents is (1) a summary of all incidents reported during the period 1 February to 29 February 1952; and (2) those incidents reported in Status Report No. 3, dated 31 January 1952, which are still pending or have been closed during the month.

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SIGHTING

DATE	TIME (Local)	LOCATION	DESCRIPTION OF INCIDENT
22 Dec 51	1030	*Columbus, Ohio	F-84 pilot observed —
20 Jan 52	1920	Spokane, Wash.	A brilliant meteor-like object was observed traveling in a 500 ft. cloud circling.
21 Jan 52	0950	*Mitchel AFB, New York	Navy TBM —
29 Jan 52	2300 and 2324	*Korea	A spherical object judged to be three feet in diameter.
29 Jan 52	1830	Falls Church, Va.	Brilliant green fireball traveling in flat arc.
10 Feb 52	Mid-day	Bandar Abbas, Iran	A shining object was observed passing over the city. explosion shook the city.
17 Feb 52	0145	Roswell, N. Mex.	Greenish-blue ball of fire was observed by a B-29 crew.
24 Feb 52	2315	Korea	B-29 navigator reported a cylindrical-shaped object with exhaust. (See Appendix IV)

SIGHTINGS OF UNIDENTIFIED OBJECTS

DESCRIPTION OF INCIDENT	LENGTH OF TIME OBSERVED	SOUND
ved —	1 1/2 Min.	None
or-like object was observed traveling at a high rate of speed under and circling.	2 Sec.	None
	2 1/2 Min.	None
et judged to be three feet in diameter.	5 Min. and 1 Min.	None
fireball traveling in flat arc.	15-30 Sec.	None
was observed passing over the city. A few minutes later, an ok the city.	—	—
all of fire was observed by a B-29 crew.	2 Sec.	None
reported a cylindrical-shaped object with some type of jet (Appendix IV)	15 Sec.	—

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SPEED	ALTITUDE	HEADING	SOURCE	ACTION OF OCCIDENTALS
Same as F-84	2,500 ft. (Est.)	285°	F-84 Pilot	See-Appendix I. Possibly a balloon.
High	Below 500 ft.	—	Four airmen	Conclusions pending.
300-500 Knots	Up to 6,000 ft.	Varied	Navy Pilot	See Appendix II. Possibly a balloon.
Same as B-29	Same as B-29	Same as B-29	Members of B-29 crews.	See Appendix III.
—	45° to 20°	225°	AF Colonel	Green fireball phenomena. No conclusions.
—	—	270°	Several Iranian Citizens	Possibly a meteor. Data was incomplete.
—	Above	135°	B-29 crew members	Green fireball phenomena. No conclusions.
Higher than B-29	Same as B-29	30°	B-29 Navigator	See Appendix IV.

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APPENDIX I

Columbus, Ohio - 22 December 1951

I. DESCRIPTION OF INCIDENT

On 22 December 1951 at 1030 EST, the pilot of an F-84 aircraft observed an unidentified object five miles east of Columbus, Ohio. The F-84 was flying at 15,000 ft. altitude on a heading of 270°. The object was first sighted at two o'clock high. It appeared to be rolling on its longitudinal axis and the shape resembled an aircraft with no tail surfaces. It was on a heading of about 90°.

The pilot observed the object for one and one half minutes during which he made a right turn in an attempt to intercept and identify the object. It was lost when the pilot turned into the sun to follow the object.

II. STATUS OF INVESTIGATION

The pilot of the F-84 was interrogated by project personnel. No new facts were brought out. It was established, however, that the pilot could have observed a balloon launched from Port Columbus Airport at about 1000 EST. The wind was 30 knots from 270° which would place the balloon in the general area of the sighting. The pilot could not pinpoint his location other than "about five miles east of Columbus".

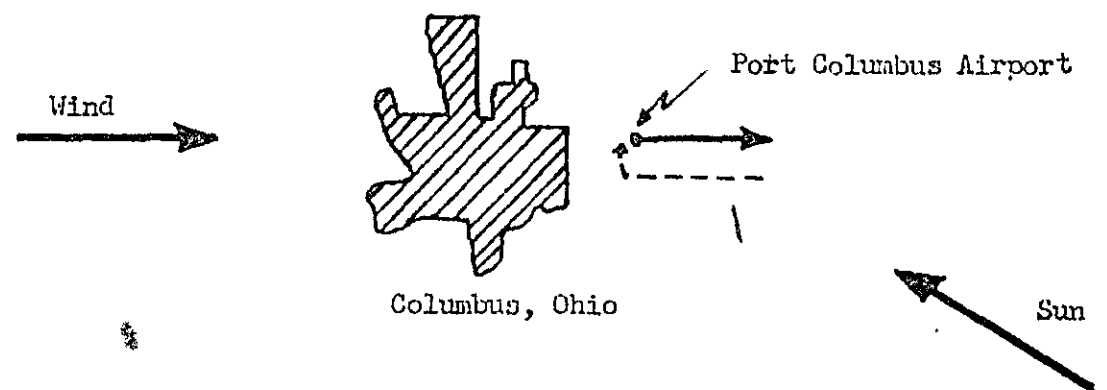
III. CONCLUSIONS

The F-84 possibly sighted a weather balloon launched from Port Columbus Airport. The reported actions of and shape of the object cannot be attributed to a balloon, however, previous reports have indicated that a balloon can be very deceptive when viewed from a high speed aircraft.

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———— Balloon Path
----- Estimated Aircraft Path

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APPENDIX II

Mitchel AFB, New York - 21 January 1952

I. DESCRIPTION OF INCIDENT

At approximately 0950 EST on 21 January 1952, a U.S. Navy pilot flying a TBM aircraft sighted an unidentified object southeast of Mitchel AFB. The TBM was on a heading at approximately 45° . When first noticed, the object was low at an angle of about 45° from the aircraft. The location of the aircraft was about three runways lengths from the end of, and lined up with, Runway #30 (300°). The object appeared to be halfway between the aircraft and the end of the runway. The pilot's first impression was that the object was a parachute and he thinks he noticed wedge or pie-shaped segregations on the top, however, he realized that the object was going cross-wind and that it could not be a drifting parachute. He judged the angular size to be the same as the angle subtended by a house on the ground and by watching the object cover the equivalent of a city block. He judged the speed to be 300 knots. He judged the altitude to be 200-300 feet. It appeared to be on a course of about 225° .

The pilot started a left turn (see overlay) in an attempt to identify the object. He states that he kept the airspeed of the TBM at about 160 knots and kept a nearly constant altitude of 6000 ft. all during the turn. He estimated that he was pulling from two to three G's in the turn. At one point near position #3 of the aircraft (see overlay) he had to increase his angle of bank to nearly 90° to keep the object from disappearing under the wing of the aircraft.

The pilot's version of the attempted interception is shown on the inclosed overlay. He stated that the paths shown are not exact due to the fact that he was concentrating on the object and not his position, although he occasionally looked at the airfield to get reference points.

The object stayed below the TBM during most of the time it was in sight. When the aircraft was somewhere near position 4, the object appeared to start a rapid climb, accelerating to an estimated 500 knots, and when it was at an angle of about 10° above the pilot's horizontal line of vision, it disappeared. When the object disappeared, the TBM was near position #5. The object did not diminish in apparent size except possibly near the end of the chase, it just disappeared. The pilot was very positive in his statement that when he was north of Mitchel AFB he could see the object. The course on which the object disappeared was established by lining up the aircraft with the apparent path of object and reading the compass.

The object appeared to be dome-shaped, or similar to the vertical cross-section of a parachute canopy. The top was light colored, "like nylon", and the under-surface was dark. It had a length to depth ratio of about 1:3.

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While the object was in level flight it appeared to oscillate with a slow period.

The total time elapsed was estimated to be about two and one half minutes.

II. STATUS OF INVESTIGATION

A field trip was made to Mitchel AFB to reinterrogate the pilot and other personnel having knowledge of facts pertaining to the sighting. The pilot's description of the incident was the same as was stated in his original report. His added details have been incorporated into Section I of this Appendix.

At 0950 EST on 21 January 1952, the weather section of Mitchel AFB launched a Rawinsonde balloon from the position shown on the overlay. These balloons are about six feet in diameter at time of launch and expand on ascending. The expansion up to 6,000 ft. can be neglected, however, as it is small. The balloons are a light gray color and have white streaks of talcum powder which is used in packing the balloons. The balloon carried a tin-foil radar 18 inch square reflector six inches below the balloon. The path of the balloon is shown on the inclosed overlay.

These balloons are tracked by radar. It was hoped that the radar operators might have recalled seeing an aircraft return circling their balloon return. They stated, however, that due to the heavy air traffic in the area, it was not uncommon to pick up aircraft returns and they did not pay any attention to them.

The tower operators on duty at the time of the sighting were interrogated. They had not seen the TBM or the balloon. The tower log showed that the first contact with the TBM was at 0955 EST at which time pilot reported sighting an object east of the field. At 1008 the pilot again called the tower to describe the incident in detail. The pilot's description was a condensed version of that given in Part I of this Appendix except he stated that the object "appeared to be a parachute canopy with a dark colored object underneath". The 0955 contact was made soon after the object was sighted, establishing the time.

III. DISCUSSION OF INCIDENT

An accurate time of the initial sighting is needed to establish the position of the balloon at this time. It is assumed that the tower clock is more accurate than the clock in the TBM, thus the time of the initial sighting was probably closer to 0954 allowing for time to contact the tower than the 0950 which was estimated by the pilot.

At 0954 the balloon would have been at about 4,000 ft. and in the position marked 4,000 ft. on the overlay. The pilot stated that the object appeared low, at an angle of 45° from vertical, and appeared to cover the

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same angle of vision as a house. This would make the slant range to the ground 8,500 ft. It can be shown that an object thought to be 30 ft. in diameter (assuming an average home is 30 ft. long) at 8,500 ft. range could also have been a six ft. diameter balloon only 1,700 ft. from the observer or at about 4,800 ft. altitude. Allowing for errors in estimation of the angle, this coincides very closely to the altitude of the balloon at 0954. The position of the balloon in respect to the ground was approximately off the end of Runway #30.

The pilot stated that the object appeared to be on a heading of 225° , the reciprocal of the heading of the TBM, and the speed of the object was about 200 knots. The balloon would appear to be traveling on a reciprocal heading and appear to be traveling at a higher rate of speed than the TBM if the pilot had assumed the balloon to be a large object close to the ground.

In examining the turn as sketched by the pilot (see overlay), it is believed that the radius of turn is too great. He stated that the air speed was kept at 160 knots and he estimated he pulled two to three G's, this would give a radius of turn of about 1,500 ft. instead of the nearly 6,000 ft. radius shown on the overlay. It will also be noted that in positions 0, 1, 2, and 3 on the overlay, the bearing of the object is relatively constant, being of about 10 o'clock from the aircraft heading. A balloon seen from an aircraft making a 360° left turn around the balloon would have a constant bearing at 9 o'clock, however, errors in the sketch of the ground tracks could account for this discrepancy.

After the position of the aircraft given as point 3 on the overlay, it is more difficult to show that the object could have been the balloon. If point 4 (of aircraft) is shifted to near point 1 (aircraft) it is possible that the pilot started another 360° turn around the balloon (see overlay).

Two major discrepancies in the theory that the object was a balloon are that the pilot was very sure that at one time during the attempt to intercept the balloon he was north or northwest of the airfield and could still see the object. In addition, shortly before the object disappeared, the line of sight of the object began to swing toward the nose of the aircraft. If this were true and the object was a balloon, the pilot should have been able to come close enough to the object to identify it as a balloon.

It should be noted that the pilot admits that the sketch could be in error. During the reinterrogation, it was brought out by the Operation's Officer at Mitchel AFB, who conducted the original interrogation, that the first sketch the pilot drew was about half the size (i.e. all radii one half) of the final sketch which has been copied in the inclosed overlay. This is further brought out by the calculations for the radius of turn. The pilot was positive that the airspeed was always 160 knots and that he was pulling about two to three G's. As stated before, this would give a radius of turn of about 1,500 ft. instead of the 6,000 ft. as shown on the overlay. A 6,000 ft. radius turn is not considered likely during any interception tactics in an aircraft as slow as a TBM. Changing the radius of the 360° turn to 1,500 ft. would "shrink" the complete sketch to one-fourth the original size (see overlay).

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The time to turn, with a 1,500 ft. radius, is 35 seconds. Assuming the turn was not a perfect circle but more of an ellipse, the time would increase to possibly 45 seconds. This is also a discrepancy since the pilot judged the elapsed time to be two and one half minutes. This is not a serious discrepancy, however, as it is known that short intervals are difficult to judge and the pilot did not actually time his maneuvers.

The description of the object could very well be that of a balloon. Observations have shown that a balloon appears to be more oval or dome-shaped than spherical and due to shadows, the bottom appears darker than the top. The talcum powder used in packing the balloon could easily give the appearance of segments such as the panels in a parachute. The oscillations of the object described by the pilot are very similar in period to those of a balloon. The pilot stated that he did not observe anything suspended from the object such as the radar reflector hanging beneath the balloon, however, the tower operator was sure that the pilot had mentioned the fact that there was something dark beneath the object when he called the tower to describe the object he had seen.

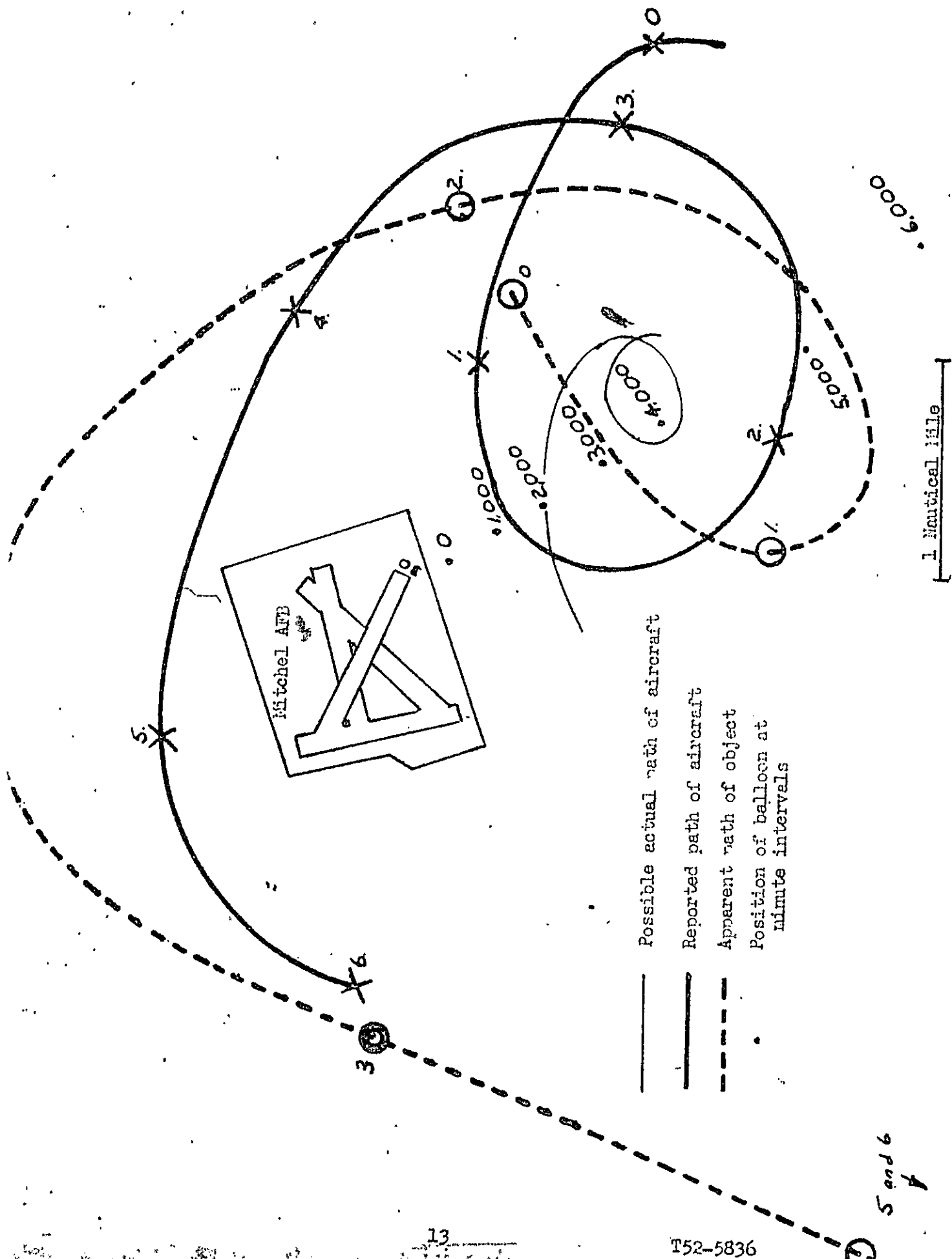
A T-11 was the only aircraft in the area near the time of the sighting. The possibility of the pilot's first seeing the balloon then the T-11 were checked but the T-11 was on an entirely different heading than that of the object, and was out of the immediate area.

IV. CONCLUSIONS

From the data obtained on this sighting, it cannot be definitely concluded that the object sighted by the TBM pilot was the Rawinsonde balloon released by the Mitchel AFB Weather Station. However, enough of the data on the reported object does correlate with that of the balloon to indicate that there is a possibility that the object observed was a balloon. There were no other reports of persons observing any unusual objects and since the object appeared to be 20 ft. to 30 ft. in diameter and very unusual in appearance at only 200 ft. to 300 ft. altitude over a thickly populated area, it would seem very likely that it would have been seen and reported by someone on the ground.

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APPENDIX III

Korea - 29 January 1952

I. DESCRIPTION OF INCIDENT

On the night of 29 January 1952, 30 miles WSW of Wonson, Korea, three members of a B-29 crew, the tail, left, and top gunner, observed a light orange colored sphere for a period of five minutes. The object was on a parallel course to the B-29 at 8 o'clock level. The color of the object was further described as being the color of the sun with an occasional bluish tint. The outer edge of the object appeared to be fuzzy and it seemed to have an internal churning movement like flames or fiery gases. The object closed in on the B-29 to an undetermined distance, and then faded away in the distance.

The aircraft was on a heading of 274°, was at 22,500 ft. altitude and was making a ground speed of 148 knots. The time of the sighting was 2300 local Korean time. The weather was CAVU.

At 2324 local Korean time, members of another B-29 crew observed an identical object near Sunchon. This object was observed for one minute. The observers were the left and tail gunners. In this instance, the B-29 was at 22,250 ft.

The sources of these reports are all World War II veterans and veterans of previous combat missions in Korea. The crews were from different squadrons and were interrogated separately.

II. DISCUSSION OF THE INCIDENT

The times that the object or objects followed the B-29's indicate that the objects were propelled by some means, which eliminates the possibility of an unguided ground-to-air missile, drop missiles, etc. The color and shape of the flame were studied by members of the ATIC Propulsion Group to determine whether or not the flame could have been the exhaust of a conventional jet engine with or without an afterburner, a pulse-jet, ram-jet, or rocket engine. None of these possibilities were considered to be applicable.

The report is somewhat similar to the reports of "fireball-fighters", a type of phenomena observed in Europe during World War II. The exact nature of this phenomena was never determined but bomber crews reported large fiery balls, similar to the sun, passing through or near their formations. There is no documented evidence or data available on this phenomena, and all the information that has been obtained is verbal from World War II bomber crewmen, consequently, few actual facts are available.

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III. CONCLUSIONS

No conclusions have been reached as to the identity or origin of these unidentified aerial objects. One possibility is that this may have been some type of flare towed by an aircraft to mark the B-29 for flak crews. No aircraft exhaust flame was reported, however.

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APPENDIX IV

Korea - 24 February 1952

I. DESCRIPTION OF INCIDENT

On the night of 24 February 1952, at 2315 local Korean time, the navigator of a B-29 observed an unidentified aerial object. The B-29 was picked up by searchlights and about 45 seconds later the navigator sighted an object approaching from 7 o'clock. The object was estimated to be over Antung at this time. The object appeared to be cylindrical in shape and had a rapidly pulsating gaseous type of exhaust trail. The exhaust trail was approximately three times the length of the object with both the tail blast and the object bluish in color.

When first observed the object appeared to be climbing at 45° , however, it leveled off approximately 4,000 ft. from the B-29 and turned toward the B-29 as if taking up an interception course. The object continued to close on the B-29 at high velocity for approximately 15 seconds after which time it broke off level flight and headed down, passing under the B-29 at a gradual angle. As the object began to descend the flame diminished in size and got brighter. The navigator estimated that the object came within 3000 ft. of the aircraft and "it appeared to be the size of an automobile". (Assumed to mean the same size as an automobile viewed from 3,000 ft.)

The B-29 was flying at 22,000 ft. altitude and at a speed of 200 knots.

The searchlights stayed with the aircraft for approximately four minutes and weak inaccurate flak was encountered during the entire incident.

Weak electronic signals were picked up by the electronic counter-measures operator for a short time.

II. STATUS OF INVESTIGATION

More details on the sighting have been requested from FEAF. Since the object resembles a surface-to-air guided missile, the incident has been referred to the ATIC guided missiles group.

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